Interacting Multiple Model Bring-Up Guide

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1 Interacting Multiple Model (IMM) with Dual-Filter Baselines

To optimize performance on the `STM32F401` hardware, an IMM filter is implemented using two parallel 7-state hypotheses ($L = 7$). This configuration exploits the numerical diversity of the Square-Root and Standard UKF formulations.

1.1 Model Hypotheses

- **Model 1 (SR-UKF):** High-precision, "Locked" tracking. Configured with $R_s = 0.1$.
- **Model 2 (Standard UKF):** High-smoothness, "Buttery" response. Configured with $R = 0.25$.

1.2 Innovation-Based Likelihood

The model selection is driven by the innovation vector $\tilde{y}$ and the innovation covariance S_{zz}. The likelihood Λ_j for each model j is:

$$\Lambda_j = \frac{1}{\sqrt{|2\pi S_{zz,j}|}} \exp\left(-0.5\tilde{y}_j^T S_{zz,j}^{-1} \tilde{y}_j\right) \tag{1}$$

2 Mode Probability and Model Likelihood

The IMM estimator selects between the "Locked" SR-UKF and "Buttery" Standard UKF based on their respective measurement likelihoods Λ_j.

2.1 Standard UKF Likelihood (Model 2)

For the full 3×3 innovation covariance $\mathbf{S}_{zz}$, the likelihood is calculated using the Gaussian density function:

$$\Lambda_2 = \frac{1}{\sqrt{(2\pi)^3 |\mathbf{S}_{zz}|}} \exp\left(-\frac{1}{2}\tilde{\mathbf{y}}^T \mathbf{S}_{zz}^{-1} \tilde{\mathbf{y}}\right) \tag{2}$$

The determinant $|\mathbf{S}_{zz}|$ is solved via Sarrus' Rule to avoid the overhead of a general matrix decomposition.

2.2 Model Probability Update

The mode probability μ_j is updated via Bayes' rule, incorporating the transition probability matrix Π to ensure temporal smoothness:

$$\mu_{j,k} = \frac{\Lambda_{j,k}\bar{c}_j}{\sum_i \Lambda_{i,k}\bar{c}_i}, \quad \bar{c}_j = \sum_i \pi_{ij}\mu_{i,k-1} \tag{3}$$

3 Model Transition Matrix (Φ)

The transition matrix Φ (or Π) defines the probability of the system switching between the SR-UKF and the Standard UKF. For a two-model system, it is defined as:

$$\Phi = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} = \begin{bmatrix} 0.98 & 0.02 \\ 0.02 & 0.98 \end{bmatrix} \tag{4}$$

Where ϕ_{ii} is the probability of remaining in the current model, and ϕ_{ij} is the probability of a mode transition. A high diagonal value (0.98) prevents "mode flickering" in the Unity visualization, ensuring a smooth blend between the "Locked" and "Buttery" states.

4 Spherical Linear Interpolation (SLERP) for IMM Mixing

Since the orientation state q exists on the S^3 manifold, the IMM mixing step must utilize *Spherical Linear Interpolation* (SLERP) rather than Euclidean averaging. This preserves the unit-norm constraint and ensures the shortest-path transition between the SR-UKF and Standard UKF hypotheses.

$$\hat{q}_{IMM} = \text{SLERP}(q_{SR}, q_{Std}, \mu_{Std}) = \frac{\sin((1-\mu)\Omega)}{\sin\Omega} q_{SR} + \frac{\sin(\mu\Omega)}{\sin\Omega} q_{Std} \tag{5}$$

where $\cos\Omega = q_{SR} \cdot q_{Std}$. This mixing ensures that the "Buttery" and "Locked" qualities are merged without introducing mid-flight normalization artifacts.

5 Innovation Normalization and Mahalanobis Distance

To compute the model likelihood Λ, the innovation vector $\tilde{\mathbf{y}}$ must be normalized by the square-root innovation covariance S_z. We solve for the normalized

innovation vector $\mathbf{v}$ via backward substitution:

$$S_z \mathbf{v} = \tilde{\mathbf{y}} \tag{6}$$

The exponent of the Gaussian likelihood (the Mahalanobis distance) is then simply the squared Euclidean norm of $\mathbf{v}$:

$$d^2 = \mathbf{v}^T \mathbf{v} = v_0^2 + v_1^2 + v_2^2 \tag{7}$$

This approach avoids the numerical cost of explicit matrix inversion while maintaining the precision of the square-root formulation.

6 IMM Context and Persistent State

To maintain the recursive Bayesian estimate of the model hypotheses, the `IMM_Context` must store the instantaneous likelihoods Λ and the integrated mode probabilities μ:

- **Likelihood (Λ_j):** The fit of model j to the current sensor frame.
- **Mode Probability (μ_j):** The long-term confidence in model j, updated via the transition matrix Φ.

$$\mu_{j,k} \propto \Lambda_{j,k} \sum_{i=1}^{M} \phi_{ij} \mu_{i,k-1} \tag{8}$$

While the Chi-Squared (χ^2) distance is calculated per-frame to derive Λ, it is not stored as a persistent state, as it represents a point-estimate of innovation error rather than a probability distribution.

7 Data Flow: Filter Likelihood to IMM Mixing

To ensure modularity, each kinematic model (CV and CA) calculates its own local likelihood based on its unique innovation $\tilde{y}$ and covariance S_z. These values are then aggregated into the `IMM_Context` for the model probability update:

$$\Lambda_{IMM} = [\text{ctx}_{SR} \to \Lambda, \quad \text{ctx}_{Std} \to \Lambda]^T \tag{9}$$

The likelihood is a point-estimate of the model's current "fit" to the physical `LSM6DSR` sensor data. Storing this value within the specific filter context allows the IMM logic to remain agnostic of the underlying filter architecture (SR-UKF vs. Standard UKF).

8 Likelihood Encapsulation and Mode Update

To maintain modularity within the IMM architecture, each filter hypothesis ($j \in \{1, 2\}$) calculates its local measurement likelihood Λ_j based on its specific innovation $\tilde{y}$ and covariance S_{zz}. These values are stored within the respective filter contexts:

$$\text{ctx}^{(j)} \rightarrow \text{likelihood} = \mathcal{N}(\tilde{y}_j; 0, S_{zz,j}) \tag{10}$$

The IMM global logic subsequently retrieves these values to update the `IMM_Context` mode probabilities μ, enabling a seamless transition between the "Locked" and "Buttery" orientation states.

9 Standard UKF Model Likelihood

The likelihood for the Standard UKF hypothesis is derived from the multivariate Gaussian distribution of the innovation residual $\tilde{\mathbf{y}}$. Given the full innovation covariance $\mathbf{S}_{zz} \in \mathbb{R}^{3\times 3}$, the likelihood Λ is defined as:

$$\Lambda = \frac{1}{\sqrt{(2\pi)^3 |\mathbf{S}_{zz}|}} \exp\left(-\frac{1}{2}\tilde{\mathbf{y}}^T \mathbf{S}_{zz}^{-1} \tilde{\mathbf{y}}\right) \tag{11}$$

For embedded execution on the `STM32F401`, the determinant $|\mathbf{S}_{zz}|$ is computed via Sarrus' Rule, and the exponent (the Mahalanobis distance) is solved using a 3×3 matrix inversion to ensure numerical parity with the SR-UKF's back-substitution method.

$$\text{Dist}_{\text{Mahalanobis}} = \tilde{\mathbf{y}}^T \mathbf{S}_{zz}^{-1} \tilde{\mathbf{y}} \tag{12}$$

10 Interacting Multiple Model (IMM) Mode Update

The final mode probability μ_j for each model $j \in \{1, 2\}$ is updated by combining the transition probabilities Φ with the current measurement likelihoods Λ.

10.1 Predicted Mode Probability

Before incorporating the new measurement, the predicted mode probability (mixing weight) $\bar{c}_j$ is calculated:

$$\bar{c}_j = \sum_{i=1}^{2} \Phi_{ij} \mu_{i,k-1} \tag{13}$$

10.2 A Posteriori Mode Update

The final probability for the current frame is normalized by the total evidence c:

$$c = \sum_{j=1}^{2} \Lambda_j \bar{c}_j, \quad \mu_{j,k} = \frac{\Lambda_j \bar{c}_j}{c} \tag{14}$$

This ensures that $\sum \mu = 1.0$, allowing the resulting probabilities to be used directly as the interpolation factor for the *Quaternion SLERP* operation.

11 IMM Initialization and Markovian Transitions

The IMM filter is initialized with a prior mode probability $\mu = [0.9, 0.1]^T$, biasing the initial estimate toward the high-precision SR-UKF. The temporal evolution of the model selection is governed by the transition matrix Φ:

$$\Phi = \begin{bmatrix} 0.98 & 0.02 \\ 0.02 & 0.98 \end{bmatrix} \tag{15}$$

A diagonal value of 0.98 represents a "High Inertia" configuration, preventing the visualization from flickering between the "Locked" and "Buttery" states during nominal sensor noise. Transitions only occur when the innovation likelihood Λ of the alternative model significantly exceeds the current hypothesis.

12 Quaternion Double-Cover and Mixing Ambiguity

The observed state $q = [1, 0, 0, 0]^T$ and 180° inversion indicate a *Double-Cover Ambiguity* failure in the IMM mixing step. Because unit quaternions q and $-q$ represent the same orientation, the individual filter hypotheses may converge to opposite poles of the S^3 manifold.

When performing `SLERP` or weighted averaging between q_{SR} and q_{Std}, if the dot product $\mathbf{d} = q_{SR} \cdot q_{Std}$ is negative, the interpolation travels the "long way" around the hypersphere. At the midpoint ($\mu = 0.5$), this results in an invalid, near-zero scalar component, which `NormalizeQuaternion` then snaps to a cardinal axis (e.g., $x = 1$), causing the "Upside Down" lock observed in Unity.

13 Synchronized Initialization and Weighting

The Interacting Multiple Model (IMM) relies on identical scaling parameters for its constituent filters to ensure a fair likelihood comparison. Both the SR-UKF and Standard UKF were initialized with the following Unscented Transformation (UT) constants:

- **Spread (α):** 0.1 (Small spread for high linearity)
- **Prior (β):** 2.0 (Optimal for Gaussian distributions)
- **Scaling (κ):** 0.0 (Standard 3D state scaling)

This synchronization ensures that the innovation covariance S_{zz} is calculated using the same sigma point distribution, allowing the `IMM_Update_Step` to accurately weight the "Locked" and "Buttery" hypotheses based solely on sensor performance rather than initialization bias.

14 Final Integration: Dual-Model IMM Performance

The implementation of the Interacting Multiple Model (IMM) filter on the `STM32F401` successfully hybridized the "Locked" precision of the SR-UKF with the "Buttery" numerical resilience of the Standard UKF. By synchronizing the Unscented Transformation parameters ($\alpha = 0.1, \beta = 2.0$) and enforcing a shortest-path *Quaternion SLERP* mixing step, the system achieved a high-fidelity orientation estimate.

14.1 Computational Efficiency

The dual-filter bank, including prediction, measurement update, likelihood calculation, and SLERP mixing, achieved the following temporal metrics:

- **Total Loop Execution:** 1.6 ms at 800 Hz ($\Delta t = 1.25$ ms scheduling).
- **CPU Overhead:** $\approx 22\%$ of the total available FPU cycles, allowing for future expansion into 16-state Constant Acceleration (CA) logic on the `STM32N6`.

14.2 Mode Probability Dynamics (μ)

The model transition matrix $\Phi = \text{diag}(0.98, 0.98)$ effectively managed the hypothesis switching. Observed behaviors include:

1. **Stationary State:** System favors Model 0 (SR-UKF) with $\mu_0 > 0.95$, resulting in zero-drift, high-stiffness orientation tracking.
2. **High-Dynamic/Impulse State:** System transitions toward Model 1 (Standard UKF) as the innovation $\tilde{y}$ exceeds the SR-UKF's strict covariance bounds, providing a smoother, low-pass filtered response to high-G noise.

15 Likelihood Saturation and Probability Stalling

A persistent mode probability of $\mu = [0.99, 0.01]^T$ indicates *Likelihood Saturation*, where the numerical ratio between Λ_1 and Λ_2 exceeds the sensitivity of the transition matrix Φ.

$$\frac{\Lambda_{SR}}{\Lambda_{Std}} \gg \frac{\phi_{11}}{\phi_{21}} \tag{16}$$

If the innovation covariance S_{zz} for the SR-UKF is extremely small (under-tuned R_s), its likelihood will mathematically dwarf the Standard UKF by several orders of magnitude (10^{15} vs 10^{-5}), preventing μ from transitioning regardless of physical motion. This is resolved by enforcing a **Likelihood Floor** and normalizing the innovation scales across both filter hypotheses.

16 Likelihood Normalization and Mode Agility

A "Stuck Mode" condition ($\mu_0 = 0.99$) occurs when the dynamic range of the Gaussian likelihoods Λ_1 and Λ_2 exceeds the numerical resolution of the transition matrix Φ. Because the likelihood is an exponential function of the Chi-Squared error, small differences in innovation result in massive discrepancies in probability density.

$$\frac{\mu_0}{\mu_1} \propto \exp\left(-\frac{1}{2}(\chi^2_{SR} - \chi^2_{Std})\right) \tag{17}$$

To restore model agility, a *Likelihood Ceiling* was implemented. By capping the maximum ratio between Λ_1 and Λ_2 at $100 : 1$, the system ensures that a significant maneuver can overcome the "sticky" transition probabilities, allowing the `IMM` to switch from the high-precision CV model to the resilient CA model in under 50 ms.

17 Model Probability Dynamics and Transition Latency

The Interacting Multiple Model (IMM) demonstrated real-time mode transitions during high-dynamic maneuvers. In a stationary state, the system converged to $\mu = [0.94, 0.06]^T$, indicating a 94% confidence in the high-precision SR-UKF (CV) hypothesis.

17.1 Innovation-Driven Mode Switching

Upon the application of external impulse (shaking), the innovation $\tilde{y}$ of the SR-UKF exceeded the 3σ threshold of its covariance S_z. This resulted in a 30% drop in μ_0, shifting the estimator's weight toward the "Buttery" Standard UKF (Model 1).

$$\Delta\mu \propto \exp\left(-0.5 \cdot \chi^2_{Std}\right) - \exp\left(-0.5 \cdot \chi^2_{SR}\right) \tag{18}$$

The transition latency $\tau \approx 25\,\text{ms}$ was sufficient to suppress high-frequency vibrational noise while maintaining the agility required for aggressive flight maneuvers.

18 Low-Inertia Model Switching ($\Phi = 0.7$)

To evaluate the sensitivity of the *Model Likelihood* (Λ) during the `STEVAL-FCU1V2` prototyping phase, the transition matrix Φ was configured for "Low Inertia."

$$\Phi = \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix} \tag{19}$$

19 Computational Benchmark and Innovation

The execution of a dual-model IMM (SR-UKF and Standard UKF) within a 1.6 ms window on an *ARM Cortex-M4* represents a significant optimization milestone. Historically, IMM architectures are reserved for *Cortex-A* or *DSP-heavy* platforms due to the $O(M \cdot L^3)$ complexity, where M is the number of models.

By utilizing the `CMSIS-DSP` library and a custom *Householder QR* implementation, this project demonstrates that high-fidelity "Model Switching" is achievable on resource-constrained hardware, providing a robust baseline for the upcoming transition to the `STM32N6` AI-accelerated platform.

This configuration allows the mode probability μ to fluctuate rapidly ($>$ 30%) in response to transient accelerometer noise. While this increases "Mode Flickering" in stationary states, it ensures that the system can transition from the high-precision CV model to the resilient CA model within a single 1.25 ms frame, providing a "Zero-Lag" response to impulsive disturbances.

20 Conclusion and Future Work

This project successfully demonstrated that a multi-model filter bank can be executed in real-time on resource-constrained ARM Cortex-M4 hardware. The transition to the `STM32N6` will leverage the existing 16-state CA model math, utilizing the N6's AXI-bus and NPU to resolve the DMA-contention and "Snail" lag observed during high-dimensional state integration.

The following tables contain the input values for our system.

Table 1: Design and Tuning Parameters for the Standard UKF (7D CV Model)

Category	**Parameter**	**Symbol**	**Value**
Scaling Parameters	Sigma Point Spread	α	0.1
	Distribution Correction	β	2.0
	Secondary Scaling	κ	0.0
Covariance Initialization	Initial Estimation Error	$\mathbf{P}_0$	0.25 (σ^2)
	Process Noise Variance	$\mathbf{Q}$	2.5×10^{-5} (σ^2_{proc})
	Measurement Noise	$\mathbf{R}$	0.5
Dimensions	State Vector Size	L	7
	Sigma Point Count	$2L + 1$	15

Table 2: Design and Tuning Parameters for the SR-UKF (7D CV Model)

Category	**Parameter**	**Symbol**	**Value**
Scaling Parameters	Sigma Point Spread	α	0.1
	Distribution Correction	β	2.0
	Secondary Scaling	κ	0.0
Cholesky Factors	Initial State Uncertainty	$\mathbf{S}_0$	0.1 (σ)
	Process Noise (Nominal)	$\mathbf{Q}_s$	0.05 (σ_{proc})
Model Complexity	State Dimension	L	7
	QR Workspace Rows	$2L + L$	21